HangulGuidedOCR: DeepSolo-style Ordered Point Guidance for Korean Apple Vision OCR

Final English Report

Abstract

This report presents HangulGuidedOCR, a Swift/Xcode library that improves Korean Apple Vision OCR outputs by applying DeepSolo-style ordered point guidance. Rather than replacing Apple Vision, the method converts Vision observations into ordered point proxies, estimates whether a Korean reading-order correction is needed, and reconstructs vertical or dispersed Korean text only when the layout evidence supports it. Across 784 ground-truth samples, raw Vision CER was 1.209 while HangulGuidedOCR(auto) reduced CER to 0.018. The result is an app-oriented correction layer for Korean posters, subway station labels, vertical signs, and stylized title layouts.

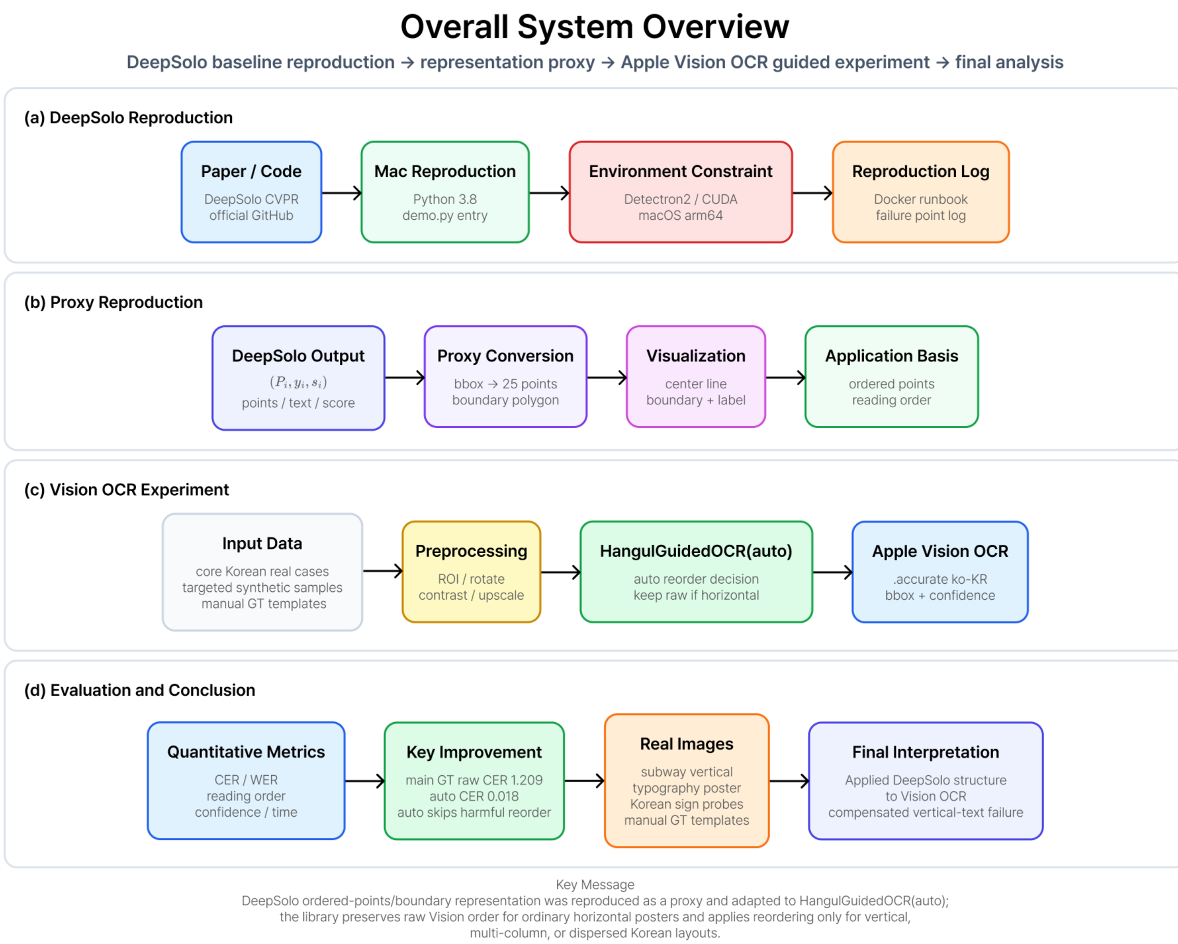


Figure 1. Overall system overview.

1. Introduction

Korean OCR in mobile apps often fails not because every character is unreadable, but because the observed boxes are not arranged in the order expected by Korean readers. Subway station boards, vertical public notices, and poster titles may be detected as fragments whose reading order is unstable. Apple Vision OCR is strong on ordinary horizontal text, so the goal is not to replace it. The goal is to add a conservative correction layer that activates only when layout evidence indicates a reading-order failure.

2. Related Work and Motivation

Scene text spotting methods such as DeepSolo model text instances with explicit point and boundary representations. This project adapts that representation idea to an Apple Vision setting: detected boxes are converted to ordered point proxies, and the proxy is used to guide Korean text reconstruction. The project also draws from OCR reading-order estimation, layout analysis, and Korean OCR evaluation, where domain-specific text order and transcription rules matter as much as character recognition.

3. DeepSolo Reproduction and Proxy

The official DeepSolo code depends on Detectron2 and CUDA-oriented inference. Because the local environment was macOS arm64, the official CUDA model inference was not claimed as fully reproduced. Instead, the experiment records the reproduction attempt separately and reproduces the DeepSolo output form at the representation level: $I \rightarrow \{(P_i, y_i, s_i)\}$. Here P_i is an ordered point sequence, y_i is the text, and s_i is confidence.

4. Method

Each Vision observation box is converted into 25 center-line points. For a box $b_i=(x_i, y_i, w_i, h_i)$, $c_{left}=(x_i, y_i+h_i/2)$, $c_{right}=(x_i+w_i, y_i+h_i/2)$, and $p_k=c_{left}+(k/24)(c_{right}-c_{left})$. The resulting point sequence provides a lightweight approximation of DeepSolo's ordered center/boundary representation.

The auto strategy first checks whether the layout is ordinary horizontal text. If so, raw Vision order is preserved. If vertical column evidence, dispersed component evidence, or title-strip evidence is strong, HangulGuidedOCR applies a Korean reading-order strategy and optional expected-order vocabulary.

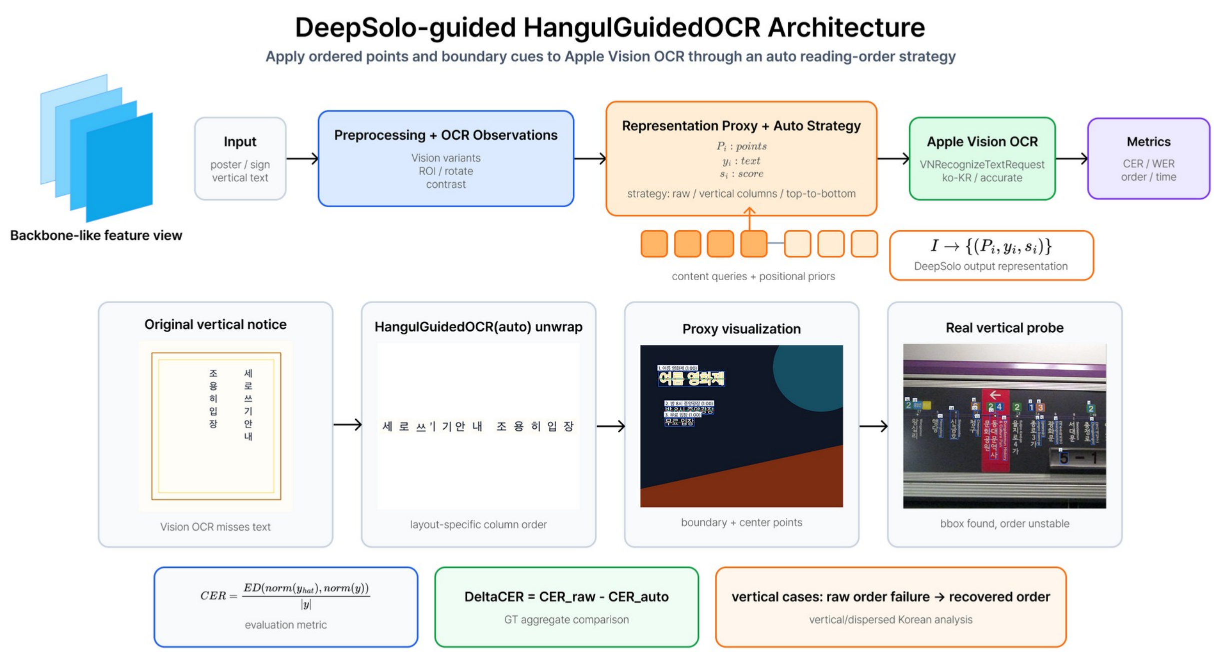


Figure 2. DeepSolo-guided HangulGuidedOCR architecture.

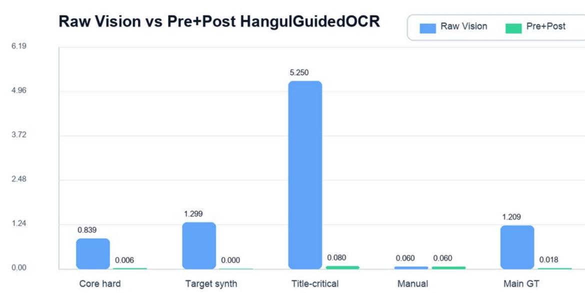
5. Experiments

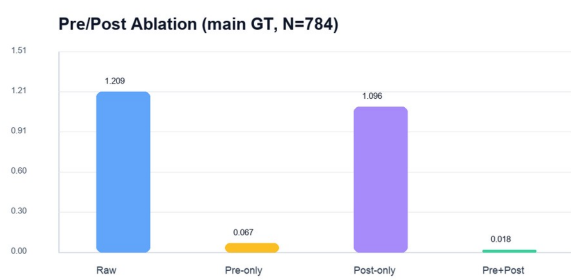
The main benchmark uses 784 ground-truth samples: 20 core hard Korean cases, 600 targeted Korean synthetic poster samples, 124 manually transcribed Canva/MiriCanvas templates, 15 public Korean sign probes, 24 title-critical local templates, and 1 public stylized spotting-failure probe. Raw Vision, preprocessing-only, postprocessing-only, and preprocessing-plus-postprocessing variants were compared.

Set	N	Raw CER	Auto CER
Main GT aggregate	784	1.209	0.018
Core real/hard Korean	20	0.835	0.005
Targeted Korean poster synthetic	600	1.202	0.000
Manual design templates	124	0.060	0.060
Public Korean sign probe	15	0.283	0.284

6. Results

The key result is that HangulGuidedOCR(auto) greatly improves layout-sensitive Korean cases while avoiding harmful reordering on ordinary horizontal templates. The aggregate CER decreased from 1.209 to 0.018. In the core hard Korean set, CER decreased from 0.835 to 0.005. In the Canva/MiriCanvas manual templates, raw Vision was already strong, and the auto strategy preserved the raw order, keeping CER at 0.060 instead of applying a harmful unconditional proxy reorder.





Stylized missing-title probe is included in Main GT and also inspected by target recall.

7. Representative Cases

The subway vertical sign demonstrates Korean column reordering. The typography poster demonstrates top-to-bottom component recovery. The stylized title case demonstrates why a pre-OCR spot proposal can be necessary when Vision does not detect the title as text.

Subway vertical sign recovery

세로 역명판 ROI를 옆으로 돌린 뒤, 검출된 두 한글 열을 한국어 역명판 읽기 순서로 재배열한다.

(a) Vertical Korean ROI

(b) Rotate 90° + ordered points

(c) Guided reading order

(d) Result

Recovered output

raw: 2 4 & Culture Park 동대문역사 문화공원

auto: 동대문역사 문화공원

target recall: 0.000 → 1.000

8. Discussion

A CER value can exceed 1.0 when a short target receives many insertion errors. Therefore, the aggregate result should be read together with per-subset variance and qualitative layout categories. Public Korean sign photos remain difficult because text may be detected but unrelated to the app's target vocabulary; ROI and expected-term priors are important in such cases.

9. Conclusion

HangulGuidedOCR translates DeepSolo's ordered point idea into a practical Swift library for Korean Apple Vision OCR correction. The method is most useful for vertical, multi-column, and dispersed Korean layouts, while its auto strategy preserves raw Vision output when the original horizontal OCR is already reliable.

References

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